Week1

```
package main

import "fmt"

func main() {
    var a, b, x, y, gcd, lcm int

    fmt.Scan(&a, &b)

    x = a
    y = b

    for y != 0 {
        rem := x % y
        x = y
        y = rem
    }

    gcd = x
    lcm = (a * b) / gcd

    fmt.Println(gcd, lcm)
}
```

Week2

```
package main

import "fmt"
```

```
func main() {
    var n int

    fmt.Println("enter the number")
    fmt.Scan(&n)

    for i := 1; i <= n; i++ {
        for j := 1; j <= n-i; j++ {
            fmt.Print(" ")
        }

        for k := 1; k <= i; k++ {
            fmt.Print(k, " ")
        }

        if i > 1 {
            fmt.Println("1")
        }

        fmt.Println()
    }
}
```

Week3

student.go:

```
package student

type Student struct {
    Name   string
    Age    int
```

```
    Rollno int
}
```

Main.go:

```
package main

import (
    "fmt"
    "student"
)

func main() {
    s := student.Student{
        Name:   "srav",
        Age:    20,
        Rollno: 22,
    }

    fmt.Println("name", s.Name)
    fmt.Println("age", s.Age)
    fmt.Println("rollno", s.Rollno)
}
```

Week4:

```
package main

import (
    "fmt"
    "math"
)
```

```
func main() {
    var num[10] float64
    var sd, mean, s float64

    for i := 1; i <= 10; i++ {
        fmt.Println("enter element:", i)
        fmt.Scan(&num[i-1])
        s += num[i-1]
    }

    mean = s / 10

    for j := 1; j < 10; j++ {
        sd += math.Pow(float64(j)-mean, 2)
    }

    sd = math.Sqrt(sd / 10)
    fmt.Println(sd)
}
```

Week5:

```
package main

import "fmt"

func main() {
    var rows int

    fmt.Print("enter the no of rows:")
    fmt.Scan(&rows)
```

```
    num := 1

    for i := 1; i <= rows; i++ {
        for j := 1; j <= i; j++ {
            fmt.Printf("%d ", num)
            num++
        }
        fmt.Println()
    }
}
```

Week6:

```
package main

import "fmt"

func main() {
    var str1, str2 string

    fmt.Print("enter first string:")
    fmt.Scan(&str1)

    fmt.Print("enter second string:")
    fmt.Scan(&str2)

    fmt.Println("concatenated string:", str1+" "+str2)
}
```

Week7:

```
package main
```

```
import "fmt"

func main() {
    var str string

    fmt.Print("enter a string:")
    fmt.Scan(&str)

    n := len(str)
    left := 0
    right := n - 1

    for left < right {
        if str[left] != str[right] {
            fmt.Println("not palindrome")
            return
        }
        left++
        right--
    }

    fmt.Println("palindrome")
}
```

Week8:

```
package main

import "fmt"
```

```
type Contact struct {
    Name    string
    Email   string
    Age     int
    Message string
}

func main() {
    var c Contact

    fmt.Println("enter the name")
    fmt.Scan(&c.Name)

    fmt.Println("enter the email")
    fmt.Scan(&c.Email)

    fmt.Println("enter the age")
    fmt.Scan(&c.Age)

    fmt.Println("enter msg")
    fmt.Scan(&c.Message)

    fmt.Println("Name:", c.Name)
    fmt.Println("Email:", c.Email)
    fmt.Println("Age:", c.Age)
    fmt.Println("Message:", c.Message)
}
```

Week9:

```
package main
```

```
import "fmt"

func main() {
    var n int
    var arr [100]float64
    var sum float64

    fmt.Print("enter no of elements:")
    fmt.Scan(&n)

    for i := 0; i < n; i++ {
        fmt.Print("enter element:")
        fmt.Scan(&arr[i])
        sum = sum + arr[i]
    }

    average := sum / float64(n)
    fmt.Printf("avg = %.2f", average)
}
```

Week10:

```
package main

import "fmt"

func main() {
    arr := [9]int{10, 20, 10, 30, 10, 0, 40, 50, 40}

    n := len(arr)
```

```
    var result [9]int
    count := 0

    for i := 0; i < n; i++ {
        found := false

        for j := 0; j < count; j++ {
            if arr[i] == result[j] {
                found = true
                break
            }
        }

        if !found {
            result[count] = arr[i]
            count++
        }
    }

    for i := 0; i < count; i++ {
        fmt.Println(result[i], " ")
    }
}
```

Week11:

```
package main

import (
    "fmt"
```

```
    "sort"
)

func main() {
    numbers := [5]int{40, 130, 5, 4, 3}
    names := [5]string{"nani", "aman", "sita", "gita", "suman"}

    sort.Sort(sort.Reverse(sort.IntSlice(numbers[:])))
    sort.Sort(sort.Reverse(sort.StringSlice(names[:])))

    fmt.Println(numbers)
    fmt.Println(names)
}
```

Week12:

```
package main

import (
    "fmt"
    "strings"
)

func main() {
    fmt.Println("strings contains:", strings.Contains("hello world, world", "world"))

    fmt.Println("Result:", strings.Count("banana", "a"))

    fmt.Println("Result:", strings.EqualFold("go", "Go"))
}
```